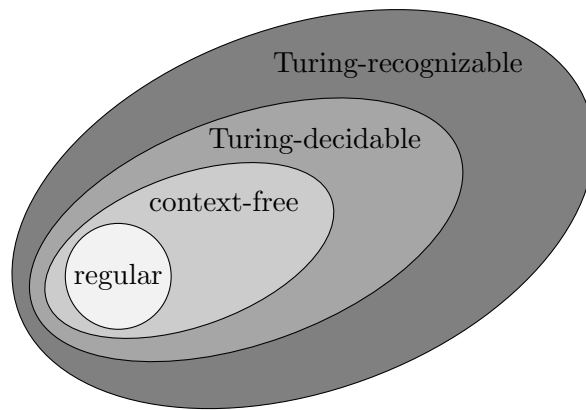


Institute of Business Administration

CSE 309: Theory of Automata

(Tentative Course Outline and Syllabus)



Institute of Business Administration

School of Mathematics and Computer Science

CSE 309: Theory of Automata

“Those who can imagine anything, can create the impossible.”

– Alan Turing

1 Logistics

	Section-1 (mega)	Section-2	Section-3
Instructors:	Shahid Hussain	Imran Rauf	
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Office hours:	TBA	TBA	TBA
Class schedule:	Tuesdays/Thursdays 4:00pm–5:15pm	Mondays/Wednesday 4:00pm–5:15pm	Fridays/Saturdays 2:30pm–3:45pm
Class location:	Auditorium	MAC-7	MTC-22

Last updated: *January 13, 2024*

2 Course Objectives

This course is about the beautiful contributions and insights that the field computer science has afforded us. We shall specifically focus on the interplay between notions of logical proof and how they link with limits on computation; which in turn has implications on what can be computed and how efficiently in the physical world. The perspective we gain by looking at our world through the computational lens is perhaps the main contribution of theoretical computer science to the human endeavor.

The course starts from basic math preliminaries, works its way through historical breakthrough results in the field and then branches off into an investigation of fundamental challenges at the frontiers of theoretical computer science. The course provides opportunities for both:

- (i) practice in developing a rigorous mathematical argument, and
- (ii) practice in developing a clear intuitive explanation of a complex argument.

During Introduction to Programming and Discrete Math students learn a new language, i.e., Discrete Math. They obtain some practice in reading, writing, speaking and listening to this new language. In a human languages class, a logical progression is to read and understand the existing classic literature in the language. Similarly, we will engage ourselves with some of the classics in theoretical computer science.

The course content is spread over three distinct modules. The first deals with what can be computed in principle: computability, the second deals with how efficiently we can compute:

complexity and finally we branch off into an investigation of fundamental challenges at the frontiers of theoretical computer science.

3 Program Learning Outcomes/Graduate Attributes

Graduate attributes (program learning outcomes - PLO's) taken from <https://www.seoulaccord.org/document.php?id=79>.

PLO-1. Academic Education

[Educational depth and breadth]

Completion of an accredited program of study designed to prepare graduates as computing professionals

PLO-2. Knowledge for Solving Computing Problems

[Breadth and depth of education and type of knowledge, both theoretical and practical]

Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements

PLO-3. Problem Analysis

[Complexity of analysis]

Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines

PLO-4. Design / Development of Solutions

[Breadth and uniqueness of computing problems, i.e., the extent to which problems are original and to which solutions have previously been identified or codified]

Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations

PLO-5. Modern Tool Usage

[Level and appropriateness of the tool to the type of activities performed]

Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations

PLO-6. Individual and Team Work

[Role in, and diversity of, the team]

Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

PLO-7. Communication

[Level of communication according to type of activities performed]

Communicate effectively with the computing community and with society at large about complex computing activities by being able to comprehend and write effective reports, design documentation, make effective presentations, and give and understand clear instructions

PLO-8. Computing Professionalism and Society

[No differentiation in this characteristic except level of practice]

Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice

PLO-9. Ethics

[No differentiation in this characteristic except level of practice]

Understand and commit to professional ethics, responsibilities, and norms of professional computing practice

PLO-10. Life-long Learning

[No differentiation in this characteristic except level of practice]

Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional

4 Course Learning Outcomes (CLO's)

Course Learning Outcome	
CLO-1	Design automata, context-free grammars, and Turing machines accepting or generating certain languages
CLO-2	Construct arguments to show that certain computational questions are uncomputable
CLO-3	Classify different computational problems in their appropriate complexity classes

4.1 CLO's to PLO's Mapping

	Academic Education	Knowledge for Solving computing Problems	Problem Analysis	Design / Development of Solutions	Modern Tool Usage	Individual and Team Work	Communication	Computing Professionalism and Society	Ethics	Life-long Learning
	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9	PLO-10
CLO-1		✓								
CLO-2			✓							
CLO-3				✓						

	CLO-1	CLO-2	CLO-3
	Design automata, context-free grammars, and Turing machines accepting or generating certain languages	Construct arguments to show that certain computational questions are uncomputable	Classify different computational problems in their appropriate complexity classes
Problem sets	✓		
Quizzes	✓	✓	
Midterm	✓	✓	
Final	✓	✓	✓

5 Course Texts

The main text for the course is

- *Introduction to the Theory of Computation* by **Michael Sipser**, 3rd edition.

Based on our choice of topics, we may refer to following texts during the course.

- *Introduction to Automata Theory, Languages, and Computation* by **John Hopcroft, Rajeev Motwani, and Jeffrey Ullman**.
- *Feynman Lectures on Computation* by **Richard P. Feynman**
- *People and Ideas in Theoretical Computer Science* (edited) by **Cristian Calude**.
- *Selected Papers on Computer Science* by **Donald Knuth**.

If necessary, additional readings will be made available online.

6 Grading Procedures

Grades will be computed as follows.

Assessment	Weight
Problem sets ($\times 3$)	$5\% \times 3 = 15\%$
Quizzes ($\times 4$)	$5\% \times 4 = 20\%$
Class participation	5%
Midterm	25%
Final	35%

6.1 Problem Sets

One of the main tools for learning the concepts we discuss in class are the problem sets. The ‘personal struggle’ you engage in with these problem sets will allow you to develop the skills

necessary for success as a theoretical computer scientist. Always spend some time thinking about these problems on your own before asking for hints, looking up solutions etc. Do not go in search of solutions online; learning the material happens when you are working on problems rather than looking up complete solutions.

You are welcome to collaborate on problem sets, provided that (1) you write up your solutions individually, and (2) you clearly cite the names of all collaborators and sources. Failure to do so will result in *zero credit*. An additional key requirement is that you should be able to explain what you submit. Inability to do so will result again in zero credit.

6.2 Quizzes

We will have five quizzes in this course and quizzes are one of the three ways to test individual and independent understanding of the contents of this course.

6.3 Exams

We will have two exams in this course (a midterm and a final). One around the mid of the semester and another (a comprehensive exam) at the end of the semester.

7 Late Work and Makeup Policy

No late solutions will be accepted and no make-up for exams or any of the quizzes will be given. If you have a valid medical excuse (for any of the quizzes, problems set, exams, etc.), the percentage of your grade corresponding to the missed work will be shifted to the final exam. Valid excuses require supporting documentation.

8 Attendance Policy

IBA attendance policy applies.

9 Academic Integrity

Each student in this course is expected to abide by the IBA Code of Conduct. Scholastic dishonesty shall be considered a serious violation of these rules and regulations and is subject to strict disciplinary action as prescribed by IBA regulations and policies. Scholastic dishonesty includes, but is not limited to, cheating on exams, plagiarism on assignments, and collusion.

- **PLAGIARISM:** Plagiarism is the act of taking the work created by another person or entity and presenting it as one's own for the purpose of personal gain or of obtaining academic credit. Plagiarism includes the submission of or incorporation of the work of others without acknowledging its provenance or giving due credit according to established academic practices. This includes the submission of material that has been appropriated, bought, received as a gift, downloaded, or obtained by any other means. Students must not, unless they have been granted permission from all faculty members concerned, submit the same assignment or project for academic credit for different courses.
- **CHEATING:** The term cheating shall refer to the use of or obtaining of unauthorized information in order to obtain personal benefit or academic credit.

- **COLLUSION:** Collusion is the act of providing unauthorized assistance to one or more person or of not taking the appropriate precautions against doing so. Any student violating academic integrity a second time in this course will receive a failing grade for the course, and additional disciplinary sanctions may be administered.

10 Course Schedule (tentative)

Week	Topic(s)	Notes
w1: [Jan 15, Jan 21)	Introduction to finite automata Preliminaries: alphabet, strings, languages	
w2: [Jan 22, Jan 28)	Finite Automata and Regular Languages Deterministic Finite Automata Designing finite automata The regular operations	
w3: [Jan 29, Feb 04)	Finite Automata and Regular Languages Nondeterministic Finite Automata Equivalence of NFAs and DFAs Closure under regular operations	Problem set 1 out Quiz 1
w4: [Feb 05, Feb 11)	Nonregular Languages Pumping lemma	
w5: [Feb 12, Feb 18)	Context-Free Grammars Designing context-free grammars Ambiguity Chomsky normal form	
w6: [Feb 19, Feb 25)	Turing Machines Definitions and examples of TMs	Quiz 2
w7: [Feb 26, Mar 03)	Variants of Turing Machines Multitape Turing machines Nondeterministic Turing machines Enumerators Equivalence with other models	Problem set 2 out
w8: [Mar 5, Mar 11]	Midterm Week	
w9: [Mar 12, Mar 18]	Spring Break	

	Week	Topic(s)	Notes
w10:	[Mar 18, Mar 24)	Decidable Languages Decidable problems concerning RL Decidable problems concerning CFL	
w11:	[Mar 25, Mar 31)	Undecidability The diagonalization method An undecidable language A Turing-unrecognizable language	Quiz 3
w12:	[Apr 01, Apr 07)	Reducibility Undecidability problems from language theory Post Correspondence Problem (PCP)	
w13:	[Apr 08, Apr 14)	Measuring Complexity Analyzing algorithms Complexity relationships among models	Problem set 3 out
w14:	[Apr 15, Apr 21)	The classes P and NP Polynomial time Polynomial time reducibility The Cook-Levin theorem Some examples	
w15:	[Apr 22, Apr 28)	NP-completeness and NP-hardness Satisfiability problem Some complete and some hard problems	Quiz 4
w16:	[Apr 29, May 05)	Space Complexity Savitch's theorem The class PSPACE PSPACE -completeness The classes L and NL	
w17:	[May 06, May 12)	Revision	
	[May 14, May 15]	Reserved Days	
	[May 16, May 27]	Final Exam	